Technology: getting the most from the experiment

Measuring team experiments

Success tips

- Set clear goals to experiment.
- Define timelines.
- Hackathon; designate winners and add to runways.
- Do not be afraid of failure.
  - Everyone experiments.
  - Mistakes are acceptable.
  - Share the big picture.
  - Celebrate success.

Challenges to success

- relying on centralized approval
- not assigning experimental value
- failing to apply continuous improvement
- responding to experimental failure with more process.

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Machine learning workflows

Gather data

- ID sources, such as logs and traces.
- Aggregate data.

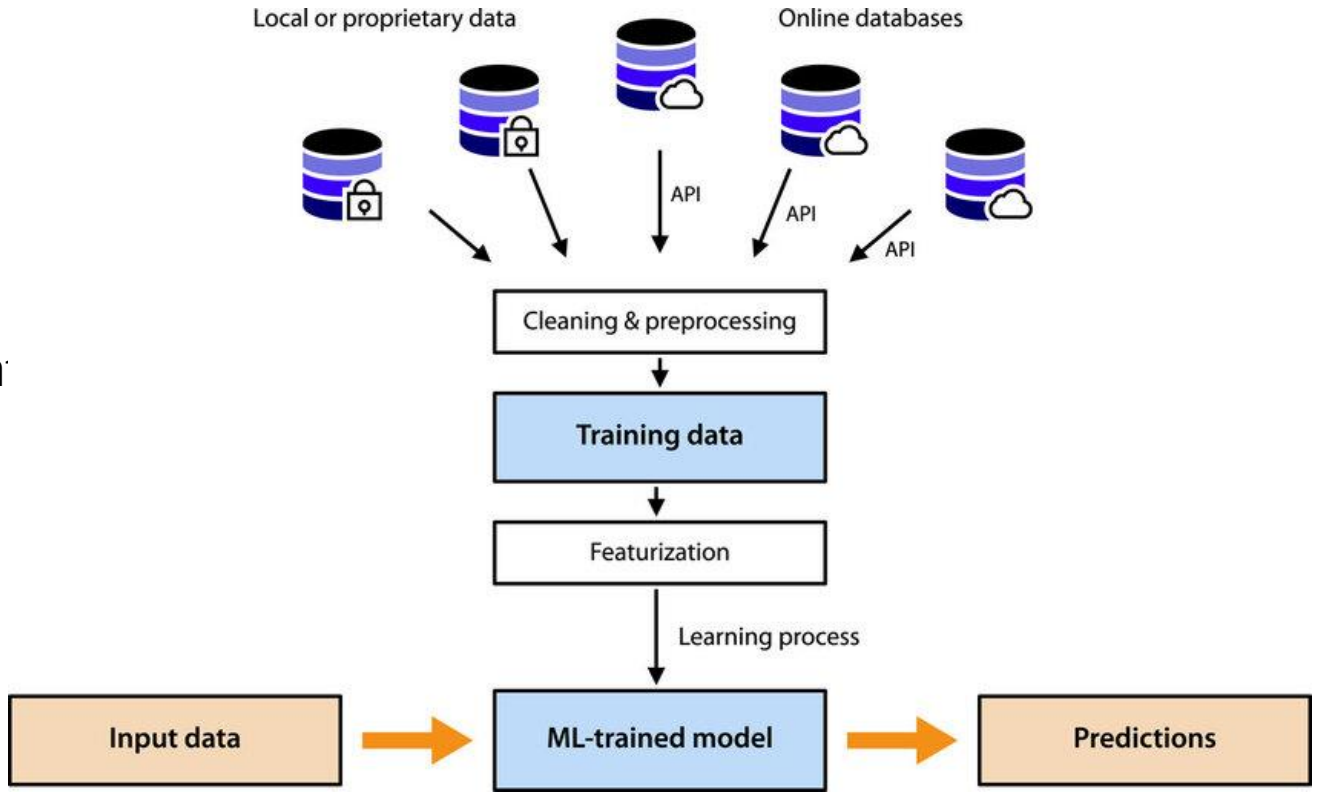
Pre-process

- Clean, verify, and format into usable set.
- Match formats, ensure reliability, de-duplica records.

Build datasets: 3 sets needed

- Training: train algorithm to process.
- Validation: estimate model accuracy.
- Test: test model performance.

Train and refine model



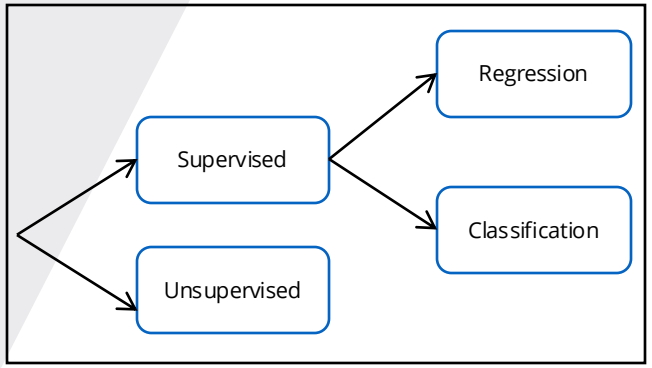
Source: https://www.researchgate.net/figure/Simplified-overview-of-a-machine-learning-workflow-The-machine-learning-model-is-trained_fig1_343440010

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Learning modules

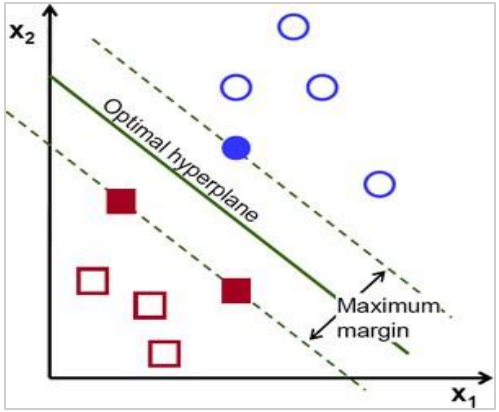
Unsupervised model

- Draw inferences and find patterns.
- Clustering: groups data points.
- Dimensionality reduction: reduces random variables by reducing features with elimination or extraction.



Source: Towardsdatascience.com

Support Vector



Source: https://www.researchgate.net/figure/Binary-SVM-thinking-The-support-vector-machine-algorithm-is-applied-to-multidimensional_fig2_319886584

Supervised model

- Model uses regression or classification.
- Regression: continuous output.
  - Linear regression: best fit line.
  - Decision tree: nodes offer two options.
  - Random forest: multiple decision trees and selects mode (middle number).
  - Neural network: network of math.
- Classification: discrete output.
  - Logistic regression: Lg scales.
  - Support vector machine
  - Hyper plane between classes
  - Naïve Bayes: What is probability of Y given X?

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Experimenting with chaos

Designed of complex, diverse, real-world issues:

- What happens when service is not accessible?
- What are results of too much load?
- Understand cascading errors.
- Determine networking shortfalls.

Some key techniques:

- Build a steady-state hypothesis.
- Simulate real-world events.
- Run in production.
- Automate and run continuously.
- Minimize blast radius.



Chaos Engineering

The discipline of experimenting on a software system in production in order to build confidence in the system's capability to withstand turbulent and unexpected conditions.

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Case story:
Implementing a reimagined DevSecOps model at a Fortune 500 Global Healthcare Company

The pilot led to significant improvements, including accelerated DevSecOps transformation, enhanced cloud adoption, a cultural shift from a project to a product mindset, reduced feedback times from months to minutes and a 50% reduction in build times.

The approach included:

- **Organizational restructuring:** dividing a large team of over 30 members into three smaller, more focused teams to enhance efficiency.
- **Code and feature management:** adopting a new branching strategy to better organize code and streamline feature development.
- **Build environment optimization:** utilizing tools like Jenkins and GitLab SaaS to improve building processes.
- **Security integration:** establishing security checkpoints to prevent the introduction of new vulnerabilities and fostering regular collaboration between security and product teams.



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